R117: Full18 Locked Results

Single-seed descriptive evidence. Historical models are not newly matched controls.

Model	Split	Top1 %	mean cost	SBS	Oracle	vs SBS %	vs Oracle %	Regret %
R117/A	VAL	71.322222	11.416106	11.427978	11.371692	-0.198173	0.389434	0.398842
R117/B	VAL	71.341975	11.416666	11.427978	11.371692	-0.194558	0.393064	0.402931
R117/A	TEST	71.446914	11.372882	11.385299	11.329932	-0.205613	0.377541	0.385552
R117/B	TEST	71.512346	11.372920	11.385299	11.329932	-0.204291	0.378864	0.387063
R94B	TEST	71.577778	11.372234	11.385299	11.329932	-0.211169	0.371939	0.379115

R117 | native/FP64 winners | TRAIN SBS | equal-task ALL | same locked weights; no TEST/LIB reselection

R117: Corrections, Harms And Cost Gains

W=cost win rate; G=mean relative cost improvement. dRegret=current-baseline; lower is better.

Comparison	Split	Corrected	Harmed	Wrong cheaper	Wrong dearer	dTop1 pp	dRegret pp	W %	G %
B vs A	VAL	1139	1123	526	505	0.019753	0.004089	2.0556	-0.005621
B vs A	TEST	1123	1070	509	516	0.065432	0.001511	2.0148	-0.003544
A vs R94B	TEST	1237	1343	602	621	-0.130864	0.006437	2.2679	-0.008328
B vs R94B	TEST	1387	1440	648	675	-0.065432	0.007948	2.5099	-0.009913

A: VAL Per-Problem Metrics

ALL equal-problem; ratios of means differ from actual per-instance regret

Problem	Top1 %	mean cost	SBS	Oracle	vs SBS %	vs Oracle %	Regret %
TSP	46.200000	8.024508	8.045354	7.990533	-0.259103	0.425191	0.415309
CVRP	48.133333	18.158969	18.269377	18.064087	-0.604334	0.525252	0.706628
ATSP	76.777778	1.638553	1.676905	1.632471	-2.287068	0.372572	0.368559
OVRP	40.488889	7.462515	7.469579	7.415777	-0.094572	0.630256	0.644097
VRPB	81.577778	9.354689	9.354876	9.318750	-0.001996	0.385672	0.384693
VRPL	37.711111	12.431974	12.437959	12.383444	-0.048124	0.391893	0.398026
VRPTW	80.400000	18.818729	18.819431	18.766139	-0.003733	0.280237	0.293198
OVRPTW	84.044444	10.141693	10.143730	10.116859	-0.020086	0.245463	0.241094
OVRPB	88.577778	6.358387	6.357254	6.341078	0.017814	0.272964	0.282050
OVRPL	41.733333	7.437910	7.445709	7.390416	-0.104751	0.642646	0.660729
VRPBL	79.466667	9.368311	9.371523	9.328366	-0.034268	0.428209	0.431706
VRPBTW	78.977778	19.944780	19.950430	19.851084	-0.028323	0.471995	0.452517
VRPLTW	79.311111	18.760947	18.765871	18.707843	-0.026237	0.283864	0.299076
OVRPBL	89.488889	6.368598	6.370093	6.352824	-0.023480	0.248292	0.259028
OVRPBTW	83.688889	10.659511	10.661098	10.623245	-0.014887	0.341386	0.308968
OVRPLTW	83.844444	10.156455	10.157423	10.129818	-0.009522	0.262962	0.267131
VRPBLTW	79.777778	19.794824	19.797044	19.706420	-0.011213	0.448605	0.435054
OVRPBLTW	83.600000	10.608551	10.609955	10.571302	-0.013235	0.352361	0.331293
ALL	71.322222	11.416106	11.427978	11.371692	-0.198173	0.389434	0.398842

R117 | native/FP64 winners | TRAIN SBS | equal-task ALL | same locked weights; no TEST/LIB reselection

A: TEST Per-Problem Metrics

ALL equal-problem; ratios of means differ from actual per-instance regret

Problem	Top1 %	mean cost	SBS	Oracle	vs SBS %	vs Oracle %	Regret %
TSP	46.622222	7.971361	7.989621	7.937593	-0.228545	0.425416	0.416395
CVRP	48.311111	18.011046	18.124983	17.918509	-0.628619	0.516436	0.674764
ATSP	76.666667	1.635427	1.674207	1.629441	-2.316340	0.367333	0.366191
OVRP	40.555556	7.446535	7.455750	7.400487	-0.123598	0.622218	0.634505
VRPB	80.911111	9.347016	9.351810	9.315298	-0.051259	0.340496	0.344562
VRPL	37.577778	12.349075	12.356138	12.300958	-0.057159	0.391164	0.399035
VRPTW	81.533333	18.683083	18.690545	18.637408	-0.039928	0.245068	0.254255
OVRPTW	84.200000	10.093571	10.093063	10.068135	0.005038	0.252641	0.244032
OVRPB	88.266667	6.368807	6.370826	6.352225	-0.031692	0.261029	0.264129
OVRPL	41.066667	7.463988	7.472542	7.418033	-0.114474	0.619494	0.629905
VRPBL	80.288889	9.393023	9.396379	9.355967	-0.035715	0.396070	0.395890
VRPBTW	78.933333	19.665213	19.665725	19.577666	-0.002602	0.447181	0.439162
VRPLTW	79.511111	18.800770	18.803136	18.745143	-0.012585	0.296754	0.315550
OVRPBL	89.488889	6.374265	6.375741	6.359310	-0.023152	0.235164	0.244380
OVRPBTW	85.066667	10.659267	10.659591	10.624748	-0.003038	0.324893	0.296206
OVRPLTW	83.266667	10.085061	10.086112	10.059001	-0.010414	0.259078	0.262461
VRPBLTW	80.111111	19.702759	19.707068	19.613284	-0.021865	0.456196	0.440914
OVRPBLTW	83.666667	10.661606	10.662149	10.625574	-0.005093	0.339106	0.317606
ALL	71.446914	11.372882	11.385299	11.329932	-0.205613	0.377541	0.385552

R117 | native/FP64 winners | TRAIN SBS | equal-task ALL | same locked weights; no TEST/LIB reselection

B: VAL Per-Problem Metrics

ALL equal-problem; ratios of means differ from actual per-instance regret

Problem	Top1 %	mean cost	SBS	Oracle	vs SBS %	vs Oracle %	Regret %
TSP	46.133333	8.025131	8.045354	7.990533	-0.251354	0.432993	0.422051
CVRP	47.422222	18.161490	18.269377	18.064087	-0.590535	0.539208	0.718075
ATSP	77.111111	1.638497	1.676905	1.632471	-2.290376	0.369174	0.362961
OVRP	41.088889	7.462221	7.469579	7.415777	-0.098511	0.626288	0.641837
VRPB	81.888889	9.353077	9.354876	9.318750	-0.019236	0.368365	0.368743
VRPL	37.844444	12.431342	12.437959	12.383444	-0.053201	0.386793	0.393362
VRPTW	80.444444	18.820675	18.819431	18.766139	0.006612	0.290612	0.304962
OVRPTW	84.066667	10.144199	10.143730	10.116859	0.004626	0.270240	0.267795
OVRPB	88.600000	6.358828	6.357254	6.341078	0.024754	0.279922	0.291019
OVRPL	41.666667	7.438389	7.445709	7.390416	-0.098323	0.649122	0.667600
VRPBL	79.377778	9.368924	9.371523	9.328366	-0.027729	0.434779	0.438698
VRPBTW	78.844444	19.945972	19.950430	19.851084	-0.022345	0.478002	0.461954
VRPLTW	79.355556	18.762647	18.765871	18.707843	-0.017183	0.292946	0.309741
OVRPBL	89.466667	6.368889	6.370093	6.352824	-0.018900	0.252884	0.265186
OVRPBTW	83.755556	10.658321	10.661098	10.623245	-0.026049	0.330184	0.297008
OVRPLTW	83.755556	10.156541	10.157423	10.129818	-0.008685	0.263802	0.268211
VRPBLTW	79.800000	19.795987	19.797044	19.706420	-0.005339	0.454506	0.438415
OVRPBLTW	83.533333	10.608866	10.609955	10.571302	-0.010267	0.355340	0.335146
ALL	71.341975	11.416666	11.427978	11.371692	-0.194558	0.393064	0.402931

B: TEST Per-Problem Metrics

ALL equal-problem; ratios of means differ from actual per-instance regret

Problem	Top1 %	mean cost	SBS	Oracle	vs SBS %	vs Oracle %	Regret %
TSP	46.911111	7.971445	7.989621	7.937593	-0.227491	0.426477	0.419635
CVRP	48.666667	18.008767	18.124983	17.918509	-0.641194	0.503715	0.669155
ATSP	76.577778	1.635473	1.674207	1.629441	-2.313599	0.370149	0.367288
OVRP	40.955556	7.445909	7.455750	7.400487	-0.131988	0.613766	0.624757
VRPB	81.133333	9.349522	9.351810	9.315298	-0.024464	0.367396	0.373584
VRPL	37.533333	12.349174	12.356138	12.300958	-0.056362	0.391965	0.399552
VRPTW	81.333333	18.685478	18.690545	18.637408	-0.027113	0.257920	0.269441
OVRPTW	84.044444	10.095993	10.093063	10.068135	0.029034	0.276696	0.268752
OVRPB	88.266667	6.368922	6.370826	6.352225	-0.029887	0.262839	0.264945
OVRPL	41.200000	7.464573	7.472542	7.418033	-0.106636	0.627390	0.639239
VRPBL	80.155556	9.392668	9.396379	9.355967	-0.039495	0.392273	0.388154
VRPBTW	79.133333	19.661791	19.665725	19.577666	-0.020002	0.429702	0.427440
VRPLTW	79.622222	18.801435	18.803136	18.745143	-0.009044	0.300306	0.316082
OVRPBL	89.422222	6.374493	6.375741	6.359310	-0.019579	0.238746	0.247415
OVRPBTW	85.000000	10.659662	10.659591	10.624748	0.000666	0.328609	0.302222
OVRPLTW	83.266667	10.083974	10.086112	10.059001	-0.021198	0.248265	0.245607
VRPBLTW	80.155556	19.703166	19.707068	19.613284	-0.019799	0.458272	0.442146
OVRPBLTW	83.844444	10.660114	10.662149	10.625574	-0.019082	0.325069	0.301721
ALL	71.512346	11.372920	11.385299	11.329932	-0.204291	0.378864	0.387063

B vs A: TEST Paired Changes

Both-wrong contributions are normalized by all problem rows; see comparison.csv for full precision.

Problem	Corrected	Harmed	Wrong cheaper	Wrong dearer	dTop1 pp	dRegret pp	W %	G %
TSP	143	130	83	75	0.288889	0.003240	5.0222	-0.004705
CVRP	186	170	133	129	0.355556	-0.005608	7.0889	0.001337
ATSP	102	106	10	12	-0.088889	0.001097	2.4889	-0.003262
OVRP	205	187	97	85	0.400000	-0.009747	6.7111	0.008172
VRPB	25	15	1	3	0.222222	0.029022	0.5778	-0.030206
VRPL	148	150	101	99	-0.044444	0.000517	5.5333	-0.000934
VRPTW	6	15	0	1	-0.200000	0.015186	0.1333	-0.015344
OVRPTW	7	14	0	0	-0.155556	0.024720	0.1556	-0.024893
OVRPB	7	7	1	3	0.000000	0.000817	0.1778	-0.001453
OVRPL	204	198	79	107	0.133333	0.009334	6.2889	-0.010407
VRPBL	20	26	2	1	-0.133333	-0.007736	0.4889	-0.000529
VRPBTW	16	7	0	0	0.200000	-0.011723	0.3556	0.010651
VRPLTW	19	14	1	0	0.111111	0.000532	0.4444	-0.003413
OVRPBL	7	10	1	1	-0.066667	0.003035	0.1778	-0.004183
OVRPBTW	2	5	0	0	-0.066667	0.006016	0.0444	-0.006449
OVRPLTW	4	4	0	0	0.000000	-0.016853	0.0889	0.009881
VRPBLTW	10	8	0	0	0.044444	0.001232	0.2222	-0.001704
OVRPBLTW	12	4	0	0	0.177778	-0.015886	0.2667	0.013645
ALL	1123	1070	509	516	0.065432	0.001511	2.0148	-0.003544

A vs R94B: TEST Paired Changes

Both-wrong contributions are normalized by all problem rows; see comparison.csv for full precision.

Problem	Corrected	Harmed	Wrong cheaper	Wrong dearer	dTop1 pp	dRegret pp	W %	G %
TSP	83	101	45	50	-0.400000	0.008510	2.8444	-0.008906
CVRP	203	210	150	184	-0.155556	0.006221	7.8444	-0.009415
ATSP	105	128	16	7	-0.511111	0.011469	2.6667	-0.013694
OVRP	268	264	116	128	0.088889	0.014194	8.5333	-0.015874
VRPB	28	35	4	1	-0.155556	-0.009821	0.7111	0.006223
VRPL	190	194	134	119	-0.088889	-0.001140	7.2000	0.000435
VRPTW	17	9	2	0	0.177778	-0.003484	0.4222	0.002873
OVRPTW	11	6	0	0	0.111111	-0.004159	0.2444	0.003815
OVRPB	10	13	2	2	-0.066667	0.006144	0.2667	-0.008123
OVRPL	229	262	126	124	-0.733333	0.009764	7.8667	-0.011324
VRPBL	31	37	4	4	-0.133333	0.027786	0.7778	-0.031584
VRPBTW	9	17	0	0	-0.177778	0.013725	0.2000	-0.014003
VRPLTW	17	16	0	1	0.022222	0.003539	0.3778	-0.007887
OVRPBL	9	10	2	1	-0.022222	0.001949	0.2444	-0.002941
OVRPBTW	8	9	0	0	-0.022222	0.011733	0.1778	-0.013980
OVRPLTW	4	4	1	0	0.000000	-0.014365	0.1111	0.008931
VRPBLTW	9	16	0	0	-0.155556	0.021846	0.2000	-0.022365
OVRPBLTW	6	12	0	0	-0.133333	0.011951	0.1333	-0.012084
ALL	1237	1343	602	621	-0.130864	0.006437	2.2679	-0.008328

B vs R94B: TEST Paired Changes

Both-wrong contributions are normalized by all problem rows; see comparison.csv for full precision.

Problem	Corrected	Harmed	Wrong cheaper	Wrong dearer	dTop1 pp	dRegret pp	W %	G %
TSP	127	132	88	80	-0.111111	0.011750	4.7778	-0.012833
CVRP	232	223	151	182	0.200000	0.000613	8.5111	-0.004798
ATSP	106	133	16	9	-0.600000	0.012567	2.6889	-0.015220
OVRP	295	273	127	134	0.488889	0.004446	9.3778	-0.006587
VRPB	31	28	3	4	0.066667	0.019201	0.7556	-0.021601
VRPL	196	202	142	121	-0.133333	-0.000623	7.5111	-0.000030
VRPTW	12	13	1	0	-0.022222	0.011702	0.2889	-0.012124
OVRPTW	9	11	0	0	-0.044444	0.020561	0.2000	-0.020953
OVRPB	13	16	1	4	-0.066667	0.006961	0.3111	-0.009427
OVRPL	257	284	112	135	-0.600000	0.019097	8.1778	-0.020772
VRPBL	32	44	3	4	-0.266667	0.020051	0.7778	-0.021051
VRPBTW	16	15	0	0	0.022222	0.002003	0.3556	-0.002875
VRPLTW	19	13	1	1	0.133333	0.004071	0.4444	-0.004576
OVRPBL	12	16	2	1	-0.088889	0.004984	0.3111	-0.006456
OVRPBTW	7	11	0	0	-0.088889	0.017749	0.1556	-0.017994
OVRPLTW	4	4	1	0	0.000000	-0.031219	0.1111	0.018975
VRPBLTW	12	17	0	0	-0.111111	0.023078	0.2667	-0.023615
OVRPBLTW	7	5	0	0	0.044444	-0.003935	0.1556	0.003507
ALL	1387	1440	648	675	-0.065432	0.007948	2.5099	-0.009913

R117: Same-Checkpoint Native LIB

TEST participated in earlier R&D; not a pristine blind test; no finish-stage reselection

Model	Split	Top1 %	mean cost	SBS	Oracle	vs SBS %	vs Oracle %	Regret %
R117/A	TSPLIB	24.489796	8.216327	8.240435	8.142167	-0.292557	0.910817	0.899313
R117/A	CVRPLIB	63.000000	67.242251	67.245364	66.945093	-0.004629	0.443884	0.590689
R117/B	TSPLIB	18.367347	8.220241	8.240435	8.142167	-0.245065	0.958882	0.998111
R117/B	CVRPLIB	67.000000	67.093199	67.245364	66.945093	-0.226283	0.221235	0.437757

R117: Provenance And Scientific Boundary

same seeded task/shuffle protocol; actual per-update sample IDs not independently archived

Item	Evidence
Logical batch	1024
Initial trainable SHA	146ee81e94bf03dd87c1e178efc41a192a3ae9348318bd26862341033e48f7b5
A checkpoint	0027a21460463258f81a0183644b3fee4807d1121852f47d28201ef3502bc84a
B checkpoint	7157ed93b116c038c0ae9565881418930079 96f671dfcf8de19c1c8c263929ae
Data version	selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018
R94B	complete:
Seed	2 only; no multi-seed significance claim
Mechanism	identical fresh tensors/shapes; random versus pretrained frozen native additions
Source of metrics	immutable saved predictions; no new forward passes or calibration